

B-ISDN signalling standards

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This paper identifies what is meant by standards and their value in the B-ISDN signalling environment. It goes on to identify the international bodies that are responsible for producing these standards and gives details of their output so far and what their future plans are expected to deliver. Consideration of the various drivers (commercial, technological, regulatory, etc) that have shaped the output and workplan is also given. In addition, there is an examination of the emerging collaborative activity and its value in limiting the potentially vast number of standards and any duplication.

1. Introduction

In any technology as complex as telecommunications, especially where there is a need for at least two realisations to be compatible to achieve their main purpose (i.e. enable effective communication), it is essential that there is a mechanism which enables the design, production and operation to be consistent. This is achieved by the use of standards comprising documents that contain agreements reached by bodies responsible for that particular aspect of telecommunications. Membership of the body will be drawn from operators, suppliers, customers and consultants with an interest in the area under study. Increasingly in the modern telecommunications world, the national/regional regulator may also be involved in the standardisation process [1]. The documents produced will be the result of considerable in-depth study, discussion and analysis enabling endorsement at company, national, regional and world level as appropriate.

While the importance of standards within the whole area of telecommunications can be gauged, it is obvious that for signalling this is particularly true. It is crucial that any signalling standard produced must be clear, comprehensive and unambiguous. In addition, signalling is there to serve the requirements of the telecommunications service being delivered, not as an end in itself. This means that the overall service required and delivered should not be modified by the signalling mechanism that supports it. These stringent requirements mean that signalling studies must be conducted with considerable care and awareness and obviously results in a process that takes some time. However, the pace of modern telecommunications developments and market needs mitigates against a lengthy process. Therefore those working in the area of signalling standards have a number of conflicting pressures that have to be satisfied so that the right standard can be produced at the right time.

Broadband ISDN (B-ISDN) is considerably more complex and has greater scope than either narrowband ISDN (N-ISDN) or PSTN. Hence from relatively simple signalling mechanisms produced to support mainly telephony-based services in the PSTN, there has been a gradual increase in the required capabilities. In N-ISDN based on the 64-kbit/s switched fabric, the scope of services was increased and, while the telephony-based, circuit-switched ancestry could still be clearly seen, the impact of both data and packet transfer requirements became increasingly significant. Now with a cell-based ATM environment the evolution of telecommunications has gone a stage further with the ability to support an even greater range of services and network capabilities. This means that there are demands on the signalling mechanism to be powerful, flexible and resilient in order to use B-ISDN to its maximum potential. To ensure what therefore must be a more complex and wide-ranging capability within the network, gaining access to the network and between networks, and achieving the greatest efficiency, incompatibility and ambiguity must be kept to a minimum. Hence the need for standardisation in B-ISDN signalling is essential.

2. Value of standards

The value of standards to those involved in telecommunications can be identified under four main headings:

- financial advantage,
- widespread acceptance,
- common understanding,
- stable environment.

The financial advantage that may be achieved by a standard approach to all aspects of telecommunications is self evident. From the supply side of industry the need to invest in customised development for each individual customer for a probable limited market is avoided. Instead, with a common basis for specification, the development can be far more generic and still satisfy a wide range of customers — hence development costs are reduced for a potentially wider market. There are also savings to be had in production where a long run of a relatively 'common' product has much greater benefit than shorter production runs of 'specialist' products. From the operators' perspective, the fact that the same product can be obtained from a large number of suppliers means that choice is increased and therefore the ability to negotiate a competitive price is increased. Further advantage is accrued in operation as the cost of running and maintaining a relatively common network fabric is less than the cost when customised builds are present in the network. In addition, procurement costs are reduced if a common specification forms the basis for the documentation to be prepared.

Widespread acceptance of product specification, operation and management procedures also has considerable benefit. For the supplier and procuring operator contractual negotiations will be greatly reduced if there is only a relatively small area of the product that has to be discussed for any limited 'operator characterisation' requirement. For operation both within and between operators' networks, acceptance of the 'agreed standards' will enable easier operation and co-operation when providing the customer with an end-to-end service. This is particularly important in the regulated multivendor environment where it is essential that a commonly agreed basis for interconnect is available [2]. In a similar way to aiding co-operation in operational matters, it is equally essential that widespread agreement is also present in the area of network management.

While benefit from cost reduction and agreement is essential, there would be little value if the standard that was produced was ambiguous. A clearly understood standard will aid the supplier in design and production. The operator will find it easier to negotiate the procurement of the product and operate and manage the network when the base standards which are used in these areas are clearly understood. Interoperation between operators will be easier when there is a commonly understood basis for interconnect. There will always be the need to strike the right balance for the standard between flexibility and service options and between being comprehensive while allowing for service differentiation.

The presence of a widely accepted and understood standard will also lead to stability in the telecommunications environment. This does not mean that there is a sterile and unmoving situation, rather that any changes/moves that are required will also have to be as a result of

common agreement. Hence any change or new standard introduced will have a well-understood and accepted basis from which it is developed rather than 'one-off' or 'quick-fix' solutions. This will aid suppliers in developing their products and the operator in the evolution of the network from understood and accepted starting points.

With the importance of signalling in the B-ISDN it can be clearly understood how the attributes of financial advantage, widespread acceptance, common understanding and stability, that a standardised approach brings, are essential and of value to supplier and operator. In addition, since this also makes for a more efficient and cost-effective service, the customer likewise benefits from B-ISDN signalling standards.

3. International standards bodies

The international bodies involved in B-ISDN signalling standards range from world-wide, through regional down to national level. There may also be an additional stage of standards activity within individual organisations and companies, but by their very nature this category will not have wide visibility and are not considered in this paper.

International standards produced by ITU-T and ETSI (and other fora) are voluntary standards unless they are identified in European (CEC) or UK law as mandatory standards. There are a number of areas of legislation under which standards have been made mandatory:

- CEC procurement directive,
- CEC terminals directive,
- CEC open network provision,
- CEC EMC directive.

The CEC directives have been made applicable to the UK via UK legislation. Hence B-ISDN signalling standards are regarded as voluntary unless of course they, in a specific instance, fall into one of the above categories.

3.1 World-wide standards makers

For B-ISDN signalling standards there are two bodies whose output is recognised world-wide, the International Telecommunications Union — Telecommunications Standardisation Sector (ITU-T) and the International Standards Organisation (ISO). Both are eminent bodies but the ITU-T has by far the greater role in studying and producing B-ISDN network standards. In addition, while not actually a 'standards body', the work and output of the ATM Forum has wide recognition in the telecommunications world and hence great impact on B-ISDN signalling studies (see section 3.3 below).

ITU-T

The ITU-T covers the whole range of telecommunications. Therefore, in order to undertake

detailed studies in the various aspects, use is made of Study Groups (SGs) which are responsible for a particular aspect and the production of resulting standards (Recommendations) if any. Within ITU-T the SG responsible for the study and production of B-ISDN signalling Recommendations is SG 11. Through its constituent Working Parties of technical experts, Recommendations covering user-to-network access, and network and transport mechanism signalling are produced as a basis for world-wide implementation by operators and suppliers. In addition, the output from ITU-T SG11 also forms the basis for study at regional, national and industry standards body level.

Within ITU-T, SG11 also receives input from other SGs in order to complete or initiate B-ISDN signalling studies; the main contributors are as follows:

- SG2 — services and numbering issues affecting B-ISDN signalling,
- SG 4 — telecommunications management,
- SG13 — aspects of network architecture (in an ATM environment) impacting on B-ISDN signalling,
- SG15 — transmission in the ATM environment,
- SG16 — multimedia requirements for B-ISDN signalling.

ISO

ISO has recently embarked upon studies leading to the production of B-ISDN signalling standards. While not involved in public telecommunications, ISO has a long history of producing standards for data communications based on inputs from the computing and private network supply industries. Hence recent studies of B-ISDN signalling by both these industries have resulted in ISO studies being initiated and the first ISO standards for B-ISDN signalling being produced.

3.2 Regional standards bodies

North America, Europe and Japan tend to be the main players in ITU-T and it is these regional organisations that are the main contributors to B-ISDN standardisation studies.

American National Standards Institute

Within the American National Standards Institute (ANSI), telecommunications is the responsibility of committee T1. ANSI T1 is responsible for the co-ordination of standards studies in N America that lead to the production of standards for use in the N American regulatory environment and the N American position for input to ITU-T standards activities. The detailed study on signalling and switching issues is conducted by the subcommittee S1. Recently ATIS (Alliance for Telecommunications Industry Solutions) has started to generate activity in the US and Caribbean in the area of

signalling and contributing to ITU-T via the US delegation. Both ANSI [3] T1 (T1 S1) and ATIS [4] undertake considerable study in the area of B-ISDN signalling which leads to ANSI standards (T1 S1) being produced.

European Telecommunication Standards Institute (ETSI)

ETSI is the European body responsible for developing telecommunications standards for use in the European Union in the harmonised market. As with the ITU-T, ETSI undertakes specific areas of telecommunications studies through specialist Technical Bodies (TBs) and Working Groups (WGs). For B-ISDN signalling the TB involved is SPS (Signalling Protocols and Switching) with the following WGs — SPS1 (internodal signalling, signalling transport mechanism), SPS3 (signalling requirements) and SPS5 (access signalling). Study in these WGs leads, as with ANSI T1, not only to a regional input (when possible) into ITU-T, but also regional standards, i.e. European Telecommunication Standards (ETs). The standards are based on the relevant ITU-T Recommendations and the ETSI target is to have the ETs as close as possible to the ITU-T Recommendation while still supporting European requirements. One aspect that the ETs do cover that is absent in the companion ITU-T Recommendations, is conformance and test specification. This important element is comprehensively covered by ETSI studies and there is a possibility that in the future this will be imported by ITU-T.

While the current method of working in ETSI will continue for some time there is now a move towards a more flexible, resilient and quicker mechanism through the use of smaller project teams to support TBs. This is in response to the needs of the telecommunications market which has imposed far more stringent time-scales on standards production than in the past. In addition ETs will become ENs (European Norms).

European Computer Manufacturers Association (ECMA)

ECMA responsibility lies in the area of private network standards. Again, using the basic output from ITU-T SG 11 on B-ISDN signalling, ECMA have produced a signalling mechanism especially tailored for the requirements of private B-ISDNs known as BQSIG. As with its narrowband counterpart, QSIG, which, while not being exactly compatible, is complementary to ETSI DSS 1, so there is a similar relationship between BQSIG and ETSI DSS 2.

However, both ETSI and ECMA realise that in the standardisation of B-ISDN signalling, the best way forward is to harmonise their future studies. Therefore there is a co-operation agreement between ECMA and ETSI for the production of signalling standards that are suitable both for the private and public network. A special joint working group, formed from ECMA and ETSI B-ISDN signalling experts, has been established to undertake this study.

ECMA have also been providing inputs to ISO (to drive those studies) and also to the ATM Forum.

Japan ITU-T co-ordination body

This body provides a forum for obtaining a common approach to standards between Japan and ITU-T. By influencing ITU-T studies in a manner complementary to the position adopted by the main Japanese players it is reasonable to assume that resulting ITU-T SG 11 B-ISDN signalling Recommendations will form the basis for the Japanese signalling standards.

3.3 *National and industry bodies*

In the era of multivendor regulated networks within a national boundary, it is now necessary to have national standards to ensure the widest choice for the customer within a seamless service. In addition, with the liberation and growth of the terminal market, there are more and more applications requiring network and signalling support. Hence there has been a growth of both national standards bodies and industry bodies representing a particular application interest.

In the UK the national standards body is the British Standards Institute (BSI). There is also the Network Interoperability Consultative Committee which, while not a standards body, does have impact on studies leading to standards being produced (see below).

Telecommunications industry bodies with an interest in B-ISDN signalling are ATM-F, DAVIC and IETF.

British Standards Institute (BSI)

BSI is the recognised national standards body for the UK. Although not directly responsible for producing the detailed B-ISDN signalling standards, it is responsible for processing the UK input into ISO and co-ordination of the UK view on the formal approval of ETSI standards.

Network Interoperability Consultative Committee (NICC)

NICC was set up under the Office of Telecommunications (OfTel) to provide advice to the Director General. One of their tasks is to ensure that standards used in the UK multivendor network satisfy the requirements of all organisations participating in telecommunications in the UK while meeting the terms of UK telecommunications regulation. A number of specialist bodies such as the Public Network Operators Interconnect Group (PNOIG) undertake studies in specific areas, such as signalling, and produce standards based on ETSI or ITU-T standards that can be used within the UK for interconnection between network operators.

ATM Forum (ATM-F) [5]

The ATM-F, while not strictly a standards body as such, is recognised as one of the main players in the production of B-ISDN signalling standards. Set up as an interest group representing mainly customer premises equipment (CPE) and private network suppliers, it has now been opened up to a wider participation and is attended by many organisations involved in ITU-T.

The purpose of the ATM-F was to sharpen the standards processing mechanism to represent both the requirements and the time-scales required in the private network environment.

The ITU-T output formed the basis for studies and modifications to suit them for private network operation. The 'standard' as initially produced by the ATM-F is a complete user/network interface (UNI) (e.g. User/Network Interface — 3.1) rather than just a signalling protocol as contained in the ITU-T base Recommendations.

ATM-F activity has now been extended into developing a private network/network interface [6] (PNNI) based on their UNI studies. In addition the ATM-F is also engaged in studies of an inter-carrier interface (B-ICI) based on the ITU Signalling System No 7, B-ISUP and UNI network interworking protocols as well as routeing mechanisms and many other aspects of telecommunications network practice.

Digital Audio-Visual Council (DAVIC) [7]

DAVIC is another experts group established to develop a common understanding of the requirements to support interactive multimedia services. Their output also ensures that the standards bodies take due account of the needs of multimedia services as they become increasingly important in the modern telecommunications environment. DAVIC has studied and produced application protocols which need to be taken into account when producing B-ISDN signalling standards. In addition, DAVIC provides another source of service requirements to develop B-ISDN signalling requirements [8].

Internet Engineering Task Force (IETF)

IETF is a relatively new experts group directed to identifying problems and providing solutions for this increasingly popular and expanding means of communication. Both public operators and suppliers recognise the importance of the Internet and ITU-T has a specific item as part of its B-ISDN signalling studies to address this topic [9].

4. Approach to signalling standardisation

Given the far-reaching impact of signalling standards in the telecommunications world it is essential that there is a widely recognised and well-understood method of producing standards. ITU and ETSI use consensus procedures wherever possible to produce their standards. Such procedures ensure the resulting standard is stable and acceptable to the widest possible community. By its very nature such a mechanism will be structured, formal and tending to the bureaucratic; it is also very time consuming. While there is an ongoing activity to minimise the inflexibility such characteristics introduce to the standards-making process, this is the price that has to be accepted for standards that must have that degree of maturity and stability to be accepted world-wide and as a basis for both contractual and regulatory commitments.

4.1 Former ITU (CCITT) approach to signalling

Until the beginning of the 1993-1996 study period the arm of the International Telecommunications Union (ITU) responsible for producing telecommunications standards (Recommendations) including signalling was the International Telephone and Telegraph Consultative Committee (CCITT). There was a fixed four-year study period allocated to the production of Recommendations by the CCITT constituent Study Groups (SG). Even if an SG were to complete and agree the technical content of a Recommendation before the study period elapsed, it could not be endorsed as a world-wide standard until approved by the full formal meeting of the CCITT (plenary assembly meeting) at the end of the study period. Similarly if a Recommendation was not completed in time to be submitted to the CCITT plenary assembly it would have to wait until the next meeting a further four years away. While in the past, given the pace of technological developments, this approach could be tolerated, it was becoming apparent by the mid to late 1980s that a new approach was required, since due to this inertia production of the required standards was lagging further and further behind the evolving technology that the telecommunications industry required.

4.2 New ITU-T approach

Being aware of the problems that the standards-making process was creating for the increasingly expanding and rapidly changing telecommunications industry the ITU decided that a new structure and mechanism was required to enable quicker delivery of standards to the 'market-place'. The World Telecommunications Standardisation Committee (WTSC) therefore decided that CCITT and its radio counterpart CCIR would be replaced by two new ITU bodies; the ITU Telecommunication Standardisation Sector (ITU-T) and the ITU Radio Standardisation Sector (ITU-R). These two bodies would again be supported by Study Groups as before but now the SGs would have far greater autonomy. This meant that rather than waiting for the full

four-year cycle to elapse before presenting candidate Recommendations to a full ITU-T meeting they would be able themselves to approve those for which they were responsible.

The process has now been refined such that when the Working Party that has undertaken the detailed study is satisfied that the draft Recommendation is sufficiently sound, mature and stable, it may put the draft forward to the SG for determination. If the SG in plenary session agrees, the document is identified to ITU members as a candidate for approval and they have until the next SG meeting to comment against it (two months before the meeting, the Telecommunications Standards Bureau (TSB) will announce if members are happy for the approvals process to go ahead). If at the next meeting there are no serious objections to the Recommendation in the determination process, it is formally approved by the SG.

4.3 Capability sets

In a similar way to the change in approvals procedure, there has also been a change in the way that B-ISDN signalling studies have been undertaken within ITU-T SG 11.

Since the studies in SG 11 are very much dependent on the architectural and technology studies relating to telecommunications networks, as undertaken by ITU-T SG 13, no signalling work could be really progressed until the SG 13 work was sufficiently mature. Within SG 13 it was realised that their studies would be extensive in scope and would therefore require a considerable time before completion was achieved. However, it was recognised that there were certain aspects that could be quickly determined and documented in Recommendations and, since there was growing market pressure for world standards, it was decided to produce these Recommendations in 'releases'. The releases would contain a complementary menu of ATM network capabilities which operators could utilise in order to deliver broadband services to their customers. This approach was accepted and therefore in Recommendation I.310 [10], SG 13 identified the network capabilities, which were required to deliver an overall broadband service, in three releases. ITU-T SG 11 examined the SG 13 Recommendation and determined the signalling requirements to support each release and decided that its studies, and the Recommendations thereby produced, would also be based on a 'release' approach.

While this was satisfactory to the relatively simple requirements of SG 13 release 1, it was realised fairly soon that the greater complexity of releases 2 and 3 meant that greater study would be required to produce the required Recommendations. With growing pressure for these signalling standards it was decided that the releases would be further partitioned to enable the more rapid production of

the necessary Recommendations. Again the partitioning was on the basis of producing complementary sets of signalling capabilities and so study of 'capability sets' (CS) was adopted by SG 11. The order in which each CS was studied was determined by the priority that had been set by the world telecommunications industry for particular broadband signalling capabilities.

4.4 Specific capabilities

The use of the capability set approach has proved very successful in partitioning and pacing studies in ITU-T SG 11 to make best use of the signalling expertise that is available. However, it was realised that even when broken down into CSs, given the pace of broadband service requirements, the time-scale for delivering signalling standards was still too long. SG 11 had already introduced, internally, the concept of 'steps' within a CS to reduce the size of each study 'bite'. This has helped ease the delay in producing the required standards, by grouping together 'families' of signalling Recommendations but even so is not considered fast enough. Therefore it has been decided that in future, as the market decides a signalling capability is required, this will be studied as a specific study item. While this may meet the time-scale constraints, there may be the danger that overall compatibility with supporting standards could be lost.

5. Current status of signalling standards

The study of signalling protocols has been under way in standards bodies since late 1990 when the ITU-T (then CCITT) SG 11 commenced work on signalling protocols to support SG 13 release 1 requirements. While there had been some initial study in bodies such as RACE [11] (Research and Development in Advanced Communications in Europe), a body of telecommunications experts that had been established by the EEC, their work acted as a stimulus to ETSI and ITU-T rather than producing actual standards. Therefore in the subsequent seven years a considerable number of signalling protocol standards have been produced and further studies undertaken. The status of signalling standards studies in the ITU-T and other bodies as at August 1997 are given in the following sections.

5.1 ITU

Brief details of the status of standards are contained in Tables 1 and 2. For full information, see the ITU Web pages [12].

5.2 ISO

Study has only recently commenced in ISO, being initiated by input of the ECMA studies which have led to the production of the first version of BQSIG (this is the

Table 1 ITU-T signalling standards currently available.

Release 1 supporting point to point bearer connections and supplementary services as standardised for support at the international signalling interface as in Recommendation Q.767 [13].	Studies completed and Recommendations in the number range Q.21##, Q.27## and Q.29## approved by ITU-T SG 11 September 1994.
Capability Set 2 Step 1 supporting additional traffic parameters (peak call rate (PCR)), connection negotiation (at establishment), connection modification (PCR), look-ahead, generic protocol for support of supplementary services and point to multipoint connections.	Studies completed and Recommendations in the number range Q.21##, Q.27## and Q.29## approved by ITU-T SG 11 May 1995 to February 1996.
Capability Set 2 Step 2 supporting frame relay in B-ISDN, call priority and additional traffic parameter (ATM) transfer capability (ATC)).	Studies completed and Recommendations in the number range Q.21##, Q.27## and Q.29## approved by ITU-T SG 11 February 1996 to January 1997.

Table 2 ITU-T signalling standards expected to be produced by 2000.

Capability Set 2 Step 1 supporting multiconnection point-to-point calls with separated bearer and call control.	Studies reasonably advanced and Recommendations in the number range Q.21##, Q.27## and Q.29## are expected to be approved by ITU-T SG 11 in May 1998.
Capability Set 2 Step 3 supporting additional traffic parameters (available bit rate (ABR), bulk transfer (ABT)), Cell Delay Variation Tolerance (CDVT) Global Tagging and Quality of Service, Connection Negotiation (PCR, Maximum Burst Size (MBS) and Sustainable Cell Rate (SCR)), Connection Modification (MBS & SCR), generic protocol (Connection Oriented Bearer Independent), Coding of DSS 2-service indicators, generic ID transport, leaf initiated join, 3rd party control, proxy signalling, light-weight signalling, generic concepts for multipoint multiconnection calls.	Studies are very advanced in some areas whereas in some aspects of CS2—3 study is yet to start. Recommendations in the number range Q.21##, Q.27## and Q.29## are expected to be approved by ITU-T SG 11 September 1997 to February 1999.
Capability Set 3 supporting multipoint multiconnection calls and IN — B-ISDN integration.	Studies have yet to start in the protocol area therefore it is unlikely that any Recommendations in the number range Q.21##, Q.27## and Q.29## will be approved by ITU-T SG 11 before late 1999.
Specific capabilities — switched virtual path (SVP), IP over switched virtual circuits, and soft PVC (permanent virtual circuits established by software control).	Studies have commenced. Recommendations in the number range Q.21##, Q.27## and Q.29## are expected to be approved by ITU-T SG 11 May 1998 to February 1999.

European Private Network Broadband Signalling System closely based on Q.2931 and the N-ISDN QSIG which is Q.931 based). ISO is endorsing the ECMA input and this will be published as an ISO standard.

In future the output of the collaboration between ETSI/ECMA on B-ISDN access signalling will also be put into ISO with the aim of having this also endorsed as an ISO standard activity, including hopefully co-operation with the ATM-Forum (see section 6).

5.3 ETSI

ETSI has been a source of considerable study and their results input to ITU-T SG11 in the area of both signalling requirements, architecture and signalling protocols. The aim of ETSI is to align the ETSI standard (ETS and EN) as closely as possible with the equivalent ITU-T Recommendation. However, there are instances when a divergence of European requirements, and therefore

signalling standards, is unavoidable. In this case an ETS is produced as a document to the equivalent Recommendation, which identifies the ETSI departures while endorsing the rest of the ITU-T document. The status of ETSs/ENs produced and under study are identified in Tables 3 and 4 — fuller details are available on the ETSI Web pages [14].

5.4 ECMA

ECMA have recently completed their initial B-ISDN signalling studies which have led to the production and approval of the initial version of the BQSIG private network signalling standard. As with its N-ISDN counterpart QSIG, BQSIG is based upon the ETSI version of the equivalent public network access signalling system. BQSIG is therefore an enhanced version of ETSI DSS 2 release 1 and CS2 step 1 ETSs/ENs with suitable additions and modifications to meet private network signalling criteria.

Table 3 ETSI signalling standards currently available.

Release 1 supporting point-to-point bearer connections.	Studies completed and ETSs approved by ETSI: Level 2: SSCOP, SSCF-UNI, SSCF: NNI Level: DSS 2 basic call, B-ISDN basic call, DSS 2/B-ISUP I/W, B-ISUP/N-ISUP I/W.
Release 1 supporting Q.767 [13] supplementary services.	Studies completed and ETSs approved by ETSI: NNI (C7):- All Release 1 supplementary services: UNI (DSS 2):- DDI, MSN, CLIP, CLIR, COLP, COLR, SUB, UUS, SSI, CUG, Cause values.
Capability Set 2 Step 1 supporting generic functional protocol and point to multipoint connections.	Studies completed and ETS approved by ETSI and out for vote NNI (C7): No ETSs straight endorsement of ITU-T SG 11 Q.27## series Recommendations: UNI (DSS 2): Point-to-multipoint and generic functional protocol.
Capability Set 2 Step 1 supporting additional traffic parameter (peak cell rate (PCR, ATC, ABR & ABT)), connection negotiation (at establishment), connection modification (PCR).	Studies completed and ETS* approved by ETSI. NNI (C7): No ETSs straight endorsement of ITU-T SG 11 Q.27## series Recommendations: UNI (DSS 2): Additional Traffic Parameters, negotiation and modification. *Under the new system of study in ETSI, standards will be produced as European Norms (ENs).

Table 4 ETSI signalling standards expected to be produced by 2000.

Capability Set 2 Step 1 supporting look ahead, pre-negotiation, separated call control, multiconnection (pt-to-pt), generic multiconnection concepts.	Studies commenced with planned ETSI approval by WG March 1998 (Note this is a joint ECMA/ETSI activity).
Capability Set 2 Step 2 supporting frame relay in B-ISDN.	Studies commenced with planned ETSI approval by WG March 1998.
Capability Set 2 Step 3 supporting additional traffic parameters (Cell Delay Variation Tolerance (CDVT) global tagging and quality of service) connection negotiation (PCR, maximum burst size (MBS) and sustainable cell rate (SCR)), connection modification (MBS and SCR), generic protocol (connection oriented bearer independent), coding of DSS 2 service indicators, generic ID transport, leaf initiated join, 3rd party control, proxy signalling, lightweight signalling, generic concepts for multipoint multiconnection calls.	ETSI studies are tracking ITU-T SG 11 with major ETSI input. ETSI endorsement or enhancement through documents will occur on completion of ITU-T study. Currently planned ETSI approval by WG March 1998.
Capability Set 3 supporting multipoint multiconnection calls and IN — B-ISDN integration.	ITU-T SG 11 studies have yet to start in the protocol area therefore it is unlikely that any Recommendations in the Q.27## and Q.29## series will be approved by ITU-T SG 11 before late 1999. Again there will be tracking of ITU-T SG 11 with major ETSI input. ETSI endorsement or enhancement through documents will occur on completion of ITU-T study. Currently planned ETSI approval by WG late 1999.
Specific capabilities — switched virtual path (SVP), IP over switched virtual circuits, and soft PVC (permanent virtual circuits established by software control).	ETSI studies are tracking ITU-T SG 11 with major ETSI input. ETSI endorsement or enhancement through documents will occur on completion of ITU-T study. Currently planned ETSI approval by WG late 1998 onwards pending completion of ITU-T SG 11 studies.

As stated in section 5.2 on ISO, the ECMA standard is now being adopted and endorsed by ISO, and for future standards there will be very close collaboration with ETSI (see section 6).

5.5 ATM Forum

The ATM-F studies started some time after ITU-T SG 11 had started work on the release 1 signalling mechanism. Therefore, in a similar fashion to ECMA, ATM-F based their initial standard on the ITU-T SG 11 access signalling work on DSS 2, but current ATM-F and ITU-T/ETSI signalling standards are not fully compatible. However, a different approach was taken by the ATM-F since the make-up of the body and its main drive was from the terminal and customer premises equipment supply side of the industry.

In addition, the ATM-F specification covered the whole interface, not just the signalling. Tables 5 and 6 list current and future ATM-F documents.

6. Future signalling standardisation activity

It can be seen from the previous section that there are a multiplicity of standards in B-ISDN signalling and the number is growing as each new interest body is established. This is bad news as far as standardisation is concerned and therefore for telecommunications in the broadband environment in general. In order to try to overcome the problem of multiple standards, convergence of studies in a number of areas are being undertaken with the aim of producing as unified a set of B-ISDN signalling standards as possible.

6.1 ETSI — ITU-T

This is perhaps the oldest form of collaboration (albeit implicit) in which ETSI, having reached an agreed position on signalling for Europe, if possible, inputs to ITU-T in order to influence the output Recommendations to reflect the European requirements. If achieved all that is required for a European standard is a simple endorsement of the ITU-T Recommendation. However, if a European complexion still has to be documented, a δ document is produced to the base ITU-T Recommendation. The aim is to make this δ document as 'thin' as possible.

6.2 ECMA — ETSI

There is now full collaborative activity by the two TB groups responsible for B-ISDN standards in ECMA and ETSI. From CS 2 step 2 onwards joint studies and meetings are being held resulting in common inputs to both ITU-T and ISO. In addition a common set of standards will be produced with minimal departures to satisfy the particular public or private network signalling requirements. There is also a joint input into ATM Forum studies in an attempt to bring all branches of B-ISDN signalling activity into as close alignment as possible.

6.3 ISO — ITU-T — ATM-F

As identified in section 5.4, with full co-operation between ETSI and ECMA in the area of B-ISDN signalling studies and standards work, and both bodies feeding the output of studies into the appropriate world-wide body, ITU-T and ISO respectively, there is a move to achieve convergence in signalling standards. In addition, there has

Table 5 ATM-F signalling documents currently available [15].

UNI V3.0 contains signalling to support the private network environment based on the ITU-T Release 2 signalling mechanism as in Rec Q.2931, plus point-to-multipoint connections and full UNI interface capabilities.	Studies completed and was the first published specification from ATM-F.
UNI V3.1 provides corrections to the V3.0 document.	Studies completed and document published providing fully implementable specification for UNI aimed at achieving interoperability with ITU-T.
UNI V4.0 signalling contains signalling to support the private network environment based on the ITU-T Release 1 plus point-to-multipoint and CS 2-1 aspects plus all UNI interface capabilities, QoS parameters, switched virtual paths and proxy signalling.	Studies completed and document published providing fully implementable specification for network UNI signalling.
P-NNI V.1 is based on UNI V4.0 but is symmetrical for use between private network nodes (also contains dynamic routing capability).	Studies completed and document published providing fully implementable specification for private network NNI.
BICI V2.0 — broadband intercarrier interface specification supports signalling based on B-ISUP.	

Table 6 ATM-F signalling documents being studied [16].

P-NNI V2 supports capabilities as in V.1 but also includes many other public network aspects of Q.2931 and ITU-T CS 2 step 1.	Studies well advanced and expected to be approved by ATM-F 4/98
AINI — enables interworking between ATM-F specification and ITU-T B-ISUP.	Studies commenced and possible completion in 1998.
BICI V3.0 — ATM-F version of B-ISUP.	Studies commenced but future of this document uncertain due to ATM-F effort on AINI.

already been an initial meeting of ITU-T and ISO and a common understanding that it is desirable to ensure this world standards convergence is reached. It is hoped then that commercially driven bodies such as the ATM-Forum will be encouraged to move their work closer to ITU-T and ISO.

6.4 Input from other bodies

While the bodies directly concerned with signalling standardisation are moving towards a more collaborative approach to their work, it is important that the 'users' of signalling are involved in this process. The aim is to have wherever possible generic signalling capabilities rather than specific signalling protocols for each and every 'user' application. Therefore within ITU-T and ETSI discussions have commenced with the appropriate groups studying IN and mobility. This is essential if future studies in B-ISDN signalling are to produce as generic a mechanism as possible. Such a generic mechanism is essential to support the wide range of services and capabilities that it is envisaged will require to be offered by operators and produced by the supply side of the industry. In addition, further input is required from bodies such as DAVIC and the IETF who represent the end-user applications that will be supported in a future B-ISDN.

7. Conclusions

This paper has attempted to show why standards are considered to be of importance in the B-ISDN signalling environment and how a number of bodies world-wide are involved in the study. It has further been illustrated how the evolving needs of world telecommunications has influenced a change in studies and led to a proliferation of standards to meet the industry's requirements. However, standardisation is not a static process and, given its importance to telecommunications, it needs to continue to develop. This is reflected in the growing collaboration between the bodies involved in the study of B-ISDN signalling. Such collaboration will hopefully increase to include all the main players involved in B-ISDN and so lead to a set of standards that has a truly world-wide application, assisting both the operators and suppliers and ultimately benefiting the telecommunications customer.

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